

Digital Audio



Audio basics

Sound perception is due to variations in air pressure near the tympanum

When an object vibrates, it produces shifts of nearby air molecules that progressively propagate and arrive at the tympanum, which in turn vibrates

Silence means no pressure variations

With respect to atmospheric pressure

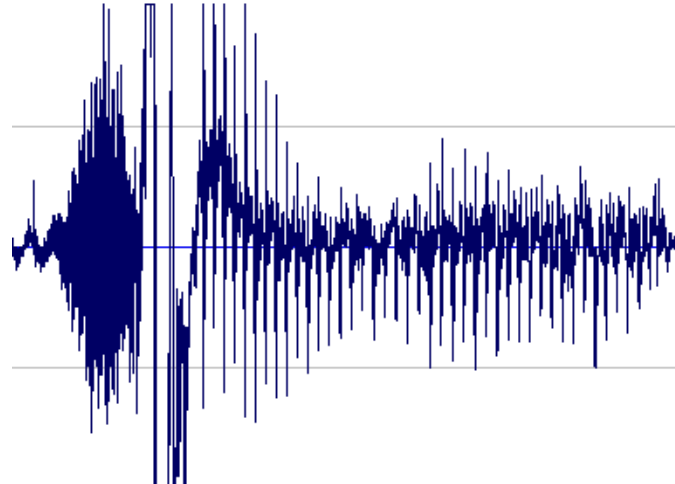
The working principle of a microphone is similar to that of the ear

A subtle membrane vibrates, producing variations in electric voltage

Sound wave

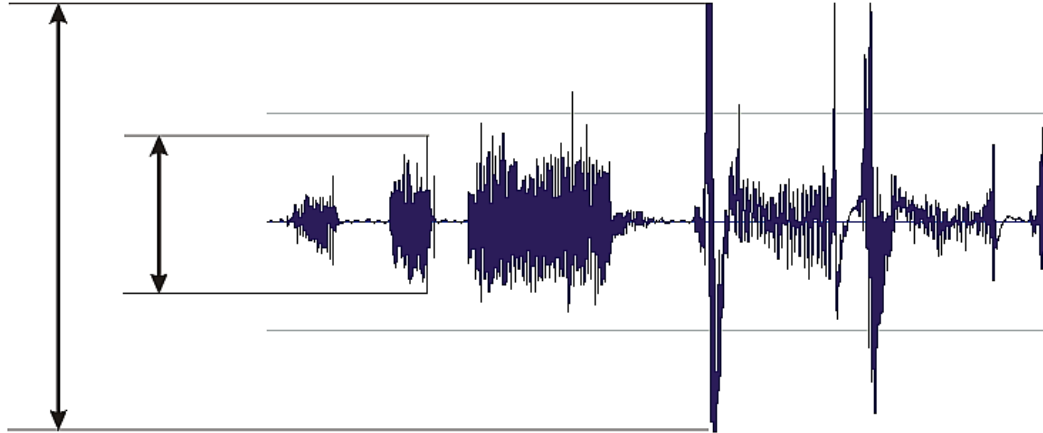
Pressure variation with time produces a *sound wave*

Example: wave
generated by the
Italian word “ciao”
 (“hello”)



Volume

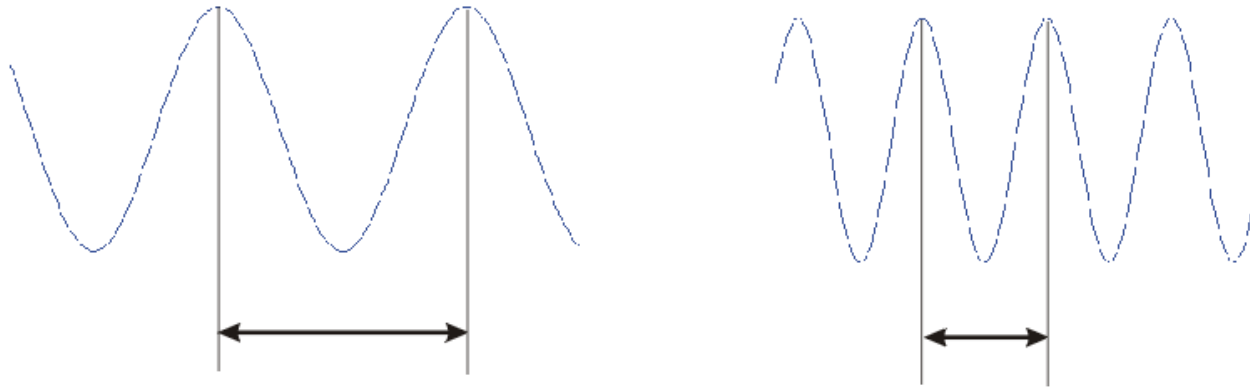
The *Volume* is given by the extent of pressure variation of the sound wave



Pitch

The *Pitch* is given by the frequency of the sound wave

How fast pressure varies...



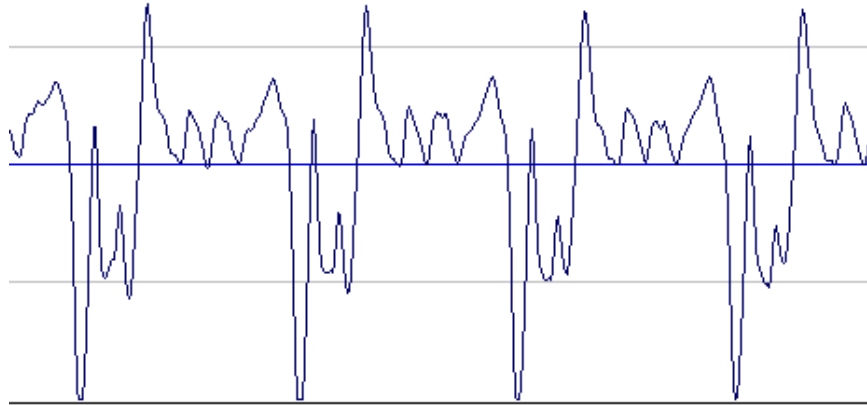
Audible frequencies for humans are normally in the range $20 \div 20,000$ Hz

Timbre

The *Timbre* is given by the “spectrum”, i.e., the set of frequencies forming the wave

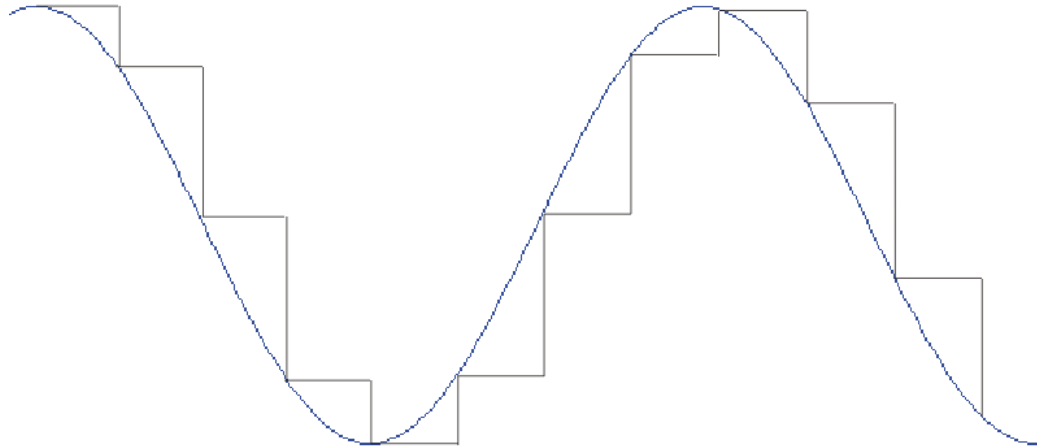
Different wave shapes produce different timbres

Example: wave generated by a saxophone



Sampling and quantization...

To be processed by means of a computer, the sound wave must be *sampled* and *quantized*



... sampling and quantization...

Sampling

Measuring the extent of the wave (its volume) at regular time intervals

The shorter the intervals, the more frequent the measures

Quantization

Finding a correspondence between each measure and a volume from a finite set of possible volumes

The bigger the set, the more accurate the volume coding

... sampling and quantization

A digitized sound is a sequence of bytes that codes its volumes

The *sampling frequency* determines the length of the time interval between consecutive samplings

Defines when the volume must be measured

The coding accuracy of volumes depends on the number of bits used in the quantization process

E.g., 8 bits → 256 possible values: all volumes will be forced to one of the 256 available possibilities

Audio digitization...

Examples of the calculation of audio file sizes

- 5 seconds of mono, non-compressed, 8-bit digital audio at 15 KHz generate a file whose size is... ?

$$5 \text{ (sec)} \cdot 15,000 \text{ (samplings per sec)} \cdot 1 \text{ (byte)} = \mathbf{75,000 \text{ bytes}}$$

- 10 seconds of stereo, non-compressed, 16-bit digital audio at 20 KHz generate a file whose size is... ?

$$10 \text{ (sec)} \cdot 2 \text{ (channels)} \cdot 20,000 \text{ (samplings per sec)} \cdot 2 \text{ (bytes)} = \mathbf{800,000 \text{ bytes}}$$

... audio digitization

Audio CD

- Sampling frequency = 44,100 Hz → volume measured 44,100 times per second
- Codable audio frequencies: from 0 to 22,050 Hz (Nyquist Theorem)
- Number of bits = 16 → the volume can assume $2^{16} = 65,536$ different values
- Stereo sound → 2 channels → $2 \cdot 16 = 32$ bits; for each second of audio → $32 \cdot 44,100$ bits = 1,764,000 bytes stored on the CD; usually an audio CD has a maximum capacity of 746.9 Mb → max 74 minutes (CD-ROMs usually have the caption “74 min / 650 Mb” because the storing of non-audio data requires redundancy bits → ~100 Mb are “lost”)



Sampling Theorem (Nyquist)

Sampling frequency ν_s

Information is not lost if ν_s is at least twice the maximum frequency component (ν_{max}) of the signal to be digitized (Fourier decomposition):

$$\nu_s \geq 2 \cdot \nu_{max}$$

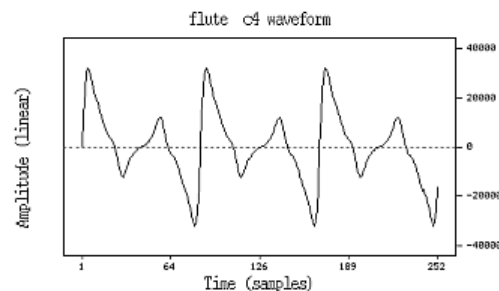
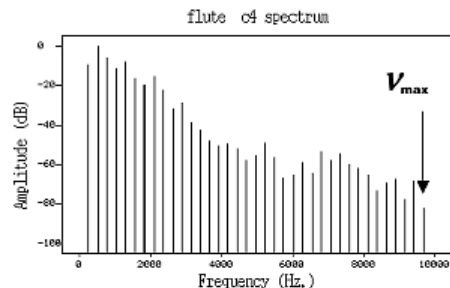
Examples

Audible frequencies: 20 Hz ÷ 20 KHz

→ Audio CD: $\nu_s = 44.1$ KHz

Voice: 100 Hz ÷ 4 KHz

→ Standard (old) telephone: $\nu_s = 8$ KHz

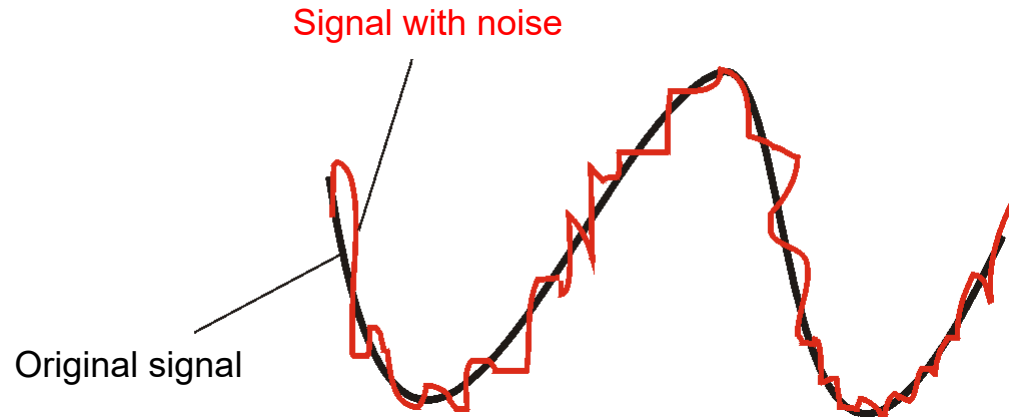


Why is digital better than analog?

(Obvious! But never to be forgotten...)

Noise affects analog signals

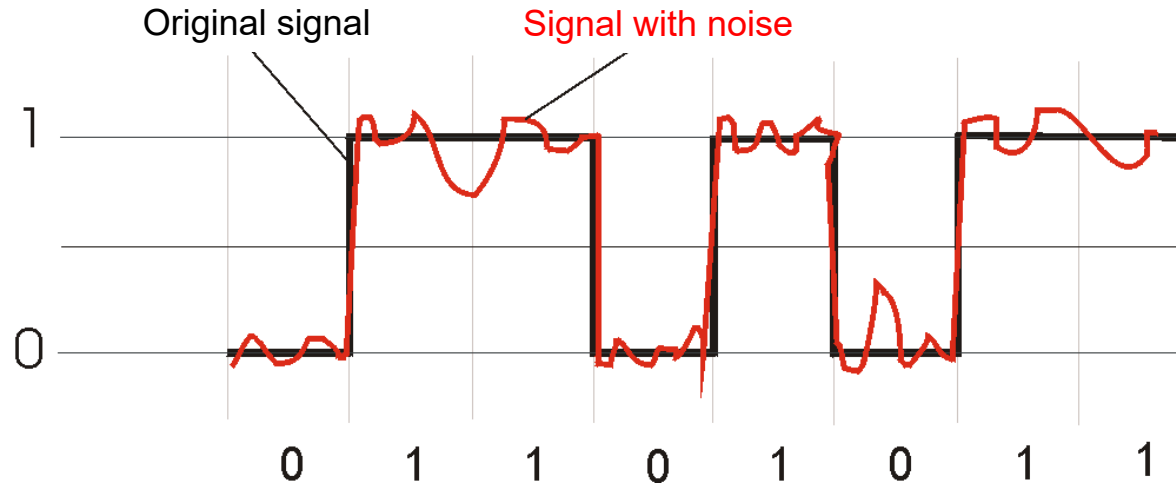
Example:



... why is digital better than analog?

Digital signals (high/low) are hardly alterable

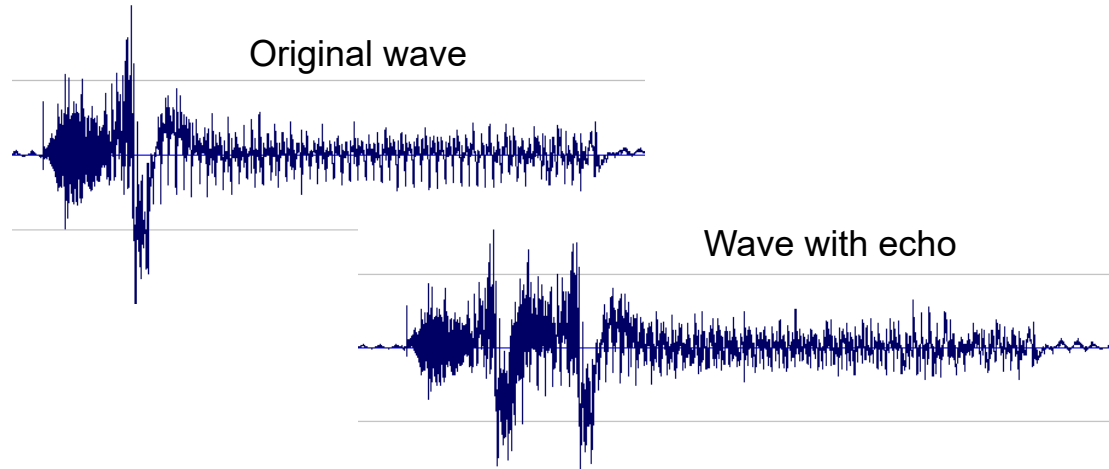
Example:



Sound editing

Several software tools allow for direct operation on sound waves

Example: echo effect



The MP3 format...

Audio compression technique

- MPEG-1 or MPEG-2 Audio Layer III
- Allows very significant reductions in file size (usually 1:4 to 1:12)
- Suitable for the Internet

Sound quality comparable to an audio CD

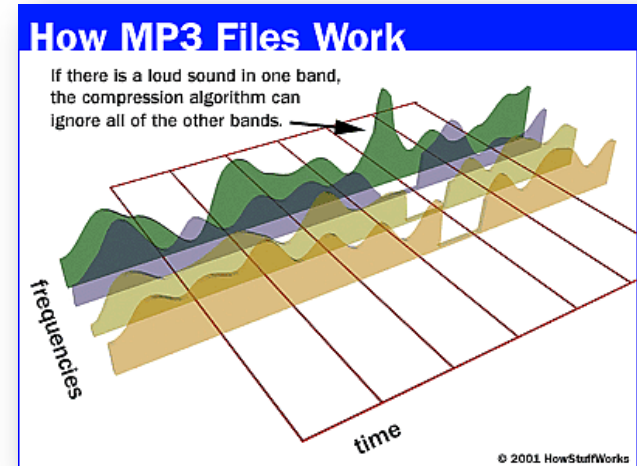
- E.g., 1 minute (MP3) may need 1 Mb, instead of the ~ 10 Mb of standard CD (it depends on the quality, however)
- Usually, 128 ÷ 320 Kbps
- Lossy data compression

... the MP3 format

Perceptual noise shaping

A compression technique based on the features of the human ear

- We cannot hear some sounds
- Some sounds are heard better than others
- If two sounds with different volumes are played at the same time, the one with the higher volume will be perceived more distinctly than the other



The AAC format

Advanced Audio Coding

- Lossy compression
- Designed to be the successor of the MP3 format
- Generally, achieves better sound quality than MP3
- Standardized by ISO and IEC, as part of the MPEG-2 and MPEG-4 specifications
- As part of the MPEG-4 standard, AAC is applicable across a broad range of digital audio environments



The Ogg Vorbis format

Ogg is the name of *Xiph.org*'s container format for audio, video, and metadata



Vorbis is the name of a specific audio compression scheme designed to be contained in *Ogg*

- The *Xiph.Org Foundation* is a non-profit corporation dedicated to protecting the foundations of Internet multimedia from control by private interests
- Roughly comparable to other formats used to store and play digital music (MP3, AAC, etc.)
- Completely free, open, and unpatented
- Usually, Vorbis files can compress to a smaller file size and still sound fine

Other common audio formats...

WAV (WAVe)

- Created by Microsoft and IBM
- Uses the PCM (Pulse Code Modulation) coding method
- Different qualities (from low to CD)
- 8/16 bits, sampling frequencies from 11,025 to 48,000 Hz, mono/stereo

AU (Sun AUdio)

- Very sophisticated
- 8/16/24/32 bits, different sampling frequencies, mono/stereo

... most common audio formats...

AIF (Audio Interchange File)

- Developed by Apple
- 8/16/24/32 bits, different sampling frequencies, mono/stereo

WMA (Windows Media Audio), RA (Real Audio)

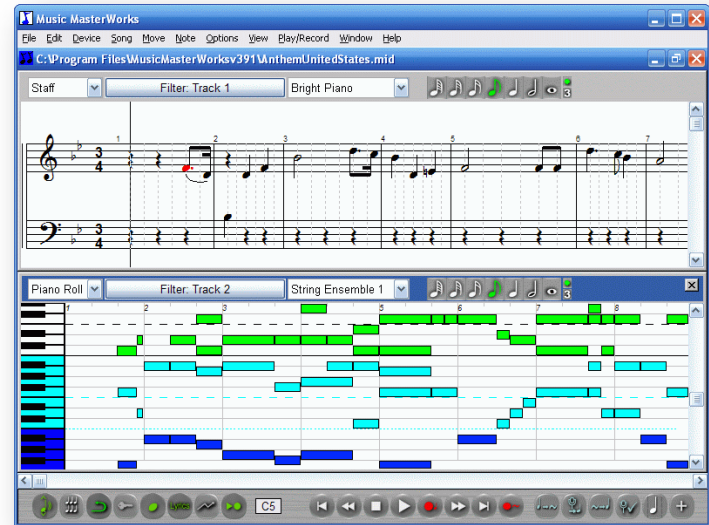
- Like MP3, compressed formats (suitable for the Internet)
- Quality is dependent on the compression level

... most common audio formats...

MIDI (Musical Instrument Digital Interface)

- **Does not contain digitized audio**
- Stores the “events” that occur during a musical performance (e.g., the note corresponding to the pressed key on a musical keyboard, “how” the key has been pressed, how long it has been kept pressed, etc.)

<https://midi.org/>



Audio formats

... most common audio formats

A variety of digital musical instruments have a MIDI interface

